

Milestone-4 Report: ExoHabitAI – Planet Habitability Prediction System

1. Introduction

ExoHabitAI is a machine learning-based web application designed to predict the habitability of exoplanets based on astrophysical parameters. The system integrates a trained ML model with a modern web interface to provide real-time predictions along with interpretability insights.

The project aims to simulate how scientists evaluate exoplanets using measurable features such as planetary mass, density, stellar temperature, and metallicity.

2. Objectives

- Predict whether a planet is **habitable or not**
 - Provide a **habitability score (probability-based)**
 - Offer **interpretable insights** based on input parameters
 - Build a **full-stack ML application** (Frontend + Backend + Model)
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3. System Architecture

The system follows a **3-tier architecture**:

1. Frontend (Client Layer)

- Built using Next.js (React framework)
- Handles user input and visualization
- Communicates with backend via API

2. Backend (Application Layer)

- Built using Flask (Python)
- Hosts ML model and prediction logic
- Handles API requests and responses

3. Machine Learning Model (Core Layer)

- XGBoost Classifier
- Trained on exoplanet dataset
- Outputs probability of habitability

4. Technologies Used

COMPONENT	TECHNOLOGY
FRONTEND	Next.js, Tailwind CSS
BACKEND	Flask (Python)
ML MODEL	XGBoost
DATA PROCESSING	NumPy, Pandas
DEPLOYMENT	Localhost (extendable to cloud)

5. Dataset & Features

The model is trained using astrophysical features:

- **pl_bmasse** → Planet mass
- **pl_dens** → Planet density
- **st_teff** → Stellar effective temperature
- **st_met** → Stellar metallicity

These features influence the possibility of sustaining life.

6. Machine Learning Model

Algorithm Used: XGBoost

XGBoost (Extreme Gradient Boosting) is chosen due to:

- High performance on structured data
 - Ability to handle non-linear relationships
 - Robustness against overfitting
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Training Process

- 1.Data preprocessing
 2. Feature selection
 - 3.Model training using XGBoost
 4. Evaluation using accuracy and probability outputs
 - 5.Model saving using joblib
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Prediction Logic

```
python
features = np.array([[pl_bmasse, pl_dens, st_teff, st_met]])
probability = model.predict_proba(features)[0][1]
```

```
prediction = 1 if probability > threshold else 0
```

7. Backend API Design

Endpoint: /predict

- **Method:** POST
- **Input (JSON):**

```
json
{
  "pl_bmasse": 100,
  "pl_dens": 2,
  "st_teff": 5500,
  "st_met": 0
}
```

- **Output:**

```
json
{
  "prediction": 1,
  "habitability_score": 0.82,
  "confidence": 0.82
}
```

8. Frontend Interface

The frontend provides:

- Interactive sliders for input
 - Real-time prediction button
 - Habitability score visualization
 - Analysis of each parameter
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9. Explainability (Analysis Module)

The system provides reasoning for predictions:

- **Mass Analysis:** Determines gravitational retention capability
- **Density Analysis:** Indicates solid vs gaseous nature
- **Temperature Analysis:** Determines habitable zone compatibility
- **Metallicity Analysis:** Indicates formation potential

This improves transparency and trust in ML predictions.

10. Challenges Faced

- Hydration errors in Next.js
 - Backend–frontend connectivity issues
 - Scaling mismatch between training and prediction
 - Model bias toward one class (habitability)
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11. Solutions Implemented

- Fixed hydration using client-side rendering control
 - Enabled CORS for API communication
 - Corrected data scaling pipeline
 - Introduced probability threshold tuning
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12. Results

- Model accuracy: ~75%
- Real-time prediction working
- UI displays both classification and score
- System provides interpretable outputs

13. Future Enhancements

- Improve accuracy using feature engineering
 - Add feature importance visualization
 - Deploy on cloud (Vercel + Render)
 - Integrate real NASA exoplanet APIs
 - Add user authentication & history tracking
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14. Conclusion

ExoHabitAI successfully demonstrates a complete ML pipeline from data to deployment. It combines machine learning with modern web technologies to create an interactive and explainable AI system for exoplanet habitability prediction.
